

Aditi Choudhary

Data Engineer

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WORK EXPERIENCE

Data Engineer | American International Group - TX, USA

May 2023 - Present

- Design and implement a scalable ETL pipeline using Apache Spark on Hadoop, processing 2TB+ of log data with PySpark and Scala, enhancing data processing efficiency by 35% through structured querying with Hive.
- Develop an optimized data pipeline on Azure, utilizing Azure Data Factory and Azure Data Lake Storage, integrating Azure Databricks to achieve 30% cost savings in storage through Spark optimizations.
- Build an analytics solution with Azure Synapse Analytics, executing complex SQL queries and integrating Power BI dashboards, improving real-time data security and accelerating decision-making by 30%.
- Optimize ETL processes on Azure Databricks, refining Spark jobs and leveraging Delta Lake partitioning, reducing data processing time by 40%.
- Streamline database operations using SQL Server Integration Services (SSIS), designing 15+ ETL packages that decreased data load times by 40%, ensuring consistency and data integrity across different sources.
- Architect high-availability Kubernetes-based data pipelines, automating deployment with Jenkins CI/CD, leading to a 40% reduction in deployment time and a 25% improvement in processing performance.

Data Engineer | Vivma Software Inc. - Hyderabad, India

Jan 2021 - Jul 2022

- Led the migration of a 15TB on-premises data warehouse to Amazon Redshift, leveraging AWS Glue and DMS for automated ETL and continuous replication, optimizing data transfer via Amazon S3, and enhancing query performance through advanced distribution techniques.
- Implemented AWS CloudWatch for real-time monitoring, improving system reliability and boosting performance by 20% through issue resolution and operational efficiency enhancements.
- Developed complex SQL procedures and views to ensure efficient data retrieval for legacy systems, enabling seamless integration into modern cloud infrastructure.
- Built a real-time data pipeline using Apache Kafka, reducing data latency by 40% by streaming external data sources into Redshift via AWS Lambda, with interactive Tableau dashboards for real-time insights.
- Designed and automated ETL workflows in Apache Airflow, integrating Snowflake and other cloud services, ensuring timely data availability through efficient DAG (Directed Acyclic Graph) scheduling.
- Optimized Snowflake data warehousing solutions, fine-tuning queries, implementing partitioning, and utilizing Snowflake Streams for incremental data loading, reducing query times by 35% and enhancing reporting efficiency.
- Developed an automation framework for property list creation, using Python and data visualization tools to analyze 100+ datasets, improving decision-making efficiency across the organization.

SKILLS

Programming Language: Python | SQL | Scala

Big Data Ecosystem: Hadoop | MapReduce | Hive | HDFS | Spark | Kubernetes | Kafka

Cloud Technologies: AWS (Glue, Kinesis, EC2, S3 Bucket, Redshift, Lambda, IAM) | MS Azure (Azure Data Lake Storage, Azure Data Factory, Azure Databricks, Azure DevOps, Azure stream analytics, Azure Synapse)

Reporting Tools: Tableau | PowerBI

Database: MS SQL Server | PostgreSQL | MySQL | Snowflake | Amazon Redshift

ETL Tools: Informatica PowerCenter | SSIS | AWS Glue | Apache Airflow

EDUCATION

Master of Engineering in Computer Science (3.96/4 CGPA)

Aug 2022 - Apr 2024

University of Cincinnati - Cincinnati, OH

Bachelor of Technology in Computer Science (3.8/4 CGPA)

Jun 2018 - May 2022

TKR College of Engineering - Hyderabad, India

CERTIFICATIONS

AWS Certified Solutions Architect - Associate

Microsoft Certified: AZ-900

Career Essentials in Generative AI by Microsoft and LinkedIn

PROJECTS

Real-Time Data Pipeline for E-Commerce Analytics

- Developed a scalable ETL pipeline using Apache Kafka, Spark, and AWS Redshift, processing 1M+ transactions daily for real-time customer insights.
- Optimized data ingestion speed by 40% through efficient partitioning and schema evolution, reducing query response time to under 2 seconds.
- Implemented data quality checks with Apache Airflow, reducing missing or incorrect records by 95%, ensuring high data integrity.

IoT Sensor Data Warehouse for Predictive Maintenance

- Built a data warehouse using Snowflake and AWS S3, storing 10TB+ of IoT sensor data from industrial machines for predictive analytics.
- Designed and automated data pipelines with Apache NiFi and AWS Lambda, reducing manual data processing time by 60%.
- Implemented machine learning-based anomaly detection, improving failure prediction accuracy by 30%, minimizing downtime in production lines.